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MACHINE LEARNING, DEEP LEARNING AND ENSEMBLE LEARNING BASED APPROACHES FOR INTRUSION DETECTION ENHANCEMENT

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Abstract. With the extensive application of Internet of Things and wireless sensor networks (WSNs) in various real-time fields, they have become increasingly vulnerable to several types of attacks that could have a serious impact in their functionalities. Recently, intrusion detection systems have become one of the crucial security components. We propose in this work an approach based on machine and deep learning for DoS and DDoS attack detection. We have evaluated, analyzed, and compared the efficiency of three learning models separately, including deep neural network (DNN), random forest, and decision tree, using the standard metrics of evaluation such as accuracy, precision, F1-score, and recall. In our contribution, an approach ensemble learning was introduced in order to enhance the accuracy rate of classification. Our study was carried out using two well-known benchmark and real-time datasets, CICIOT-2023 and WSN-DS, intended for wireless sensor networks and IoT. These datasets containing various types of attacks. CICIOT2023 dataset contains 33 attacks divided into 7 classes, while WSN-DS dataset contains four types of DoS attacks, including Blackhole, Grayhole, Flooding, and TDMA. The experiment result demonstrate the effectiveness of our approach in attacks detection with high accuracy achieved close to perfect. Our approach result demonstrates that using the ensemble learning technique works better than each ML architecture independently.

INTRODUCTION

Internet of Things (IoT) technology has made our lives connected, simple, and smart by integrating physical objects to the Internet in various fields in order to transmit data to users using a wireless sensor networks (WSNs) (Salmi & Oughdir, 2023). Unfortunately,

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these sensors have limited resources in terms of memory, CPU, battery, etc. (Nedham & Al-Qurabat, 2023). Sensors in the network can produce huge amounts of data and may have heterogeneous features or similar devices and are referred to as homogeneous. The applications of IoT and WSN have become diverse and are increasing day by day (Akyildiz & Vuran, 2010; Zemrane et al., 2018), and since they are interconnected with each other via the Internet and even deployed in harsh environments, they are vulnerable to many types of attacks. Among these are DoS attacks and DDoS attacks that could have a serious impact on WSNs functionalities. These types of attacks are carried out using the installation of malware applications called botnets (Bhushan et al., 2018; Sarraf, 2020; Weiss, 2019) and are aimed at specific targets, such as websites, online platforms, or services. They are launched from a single Internet connection point with the intention of flooding it with more traffic than it can manage, as illustrated in Figure 1. Their main negative impact is to interrupt the normal functioning of a server, service, or network by saturating it with a flood of Internet traffic.

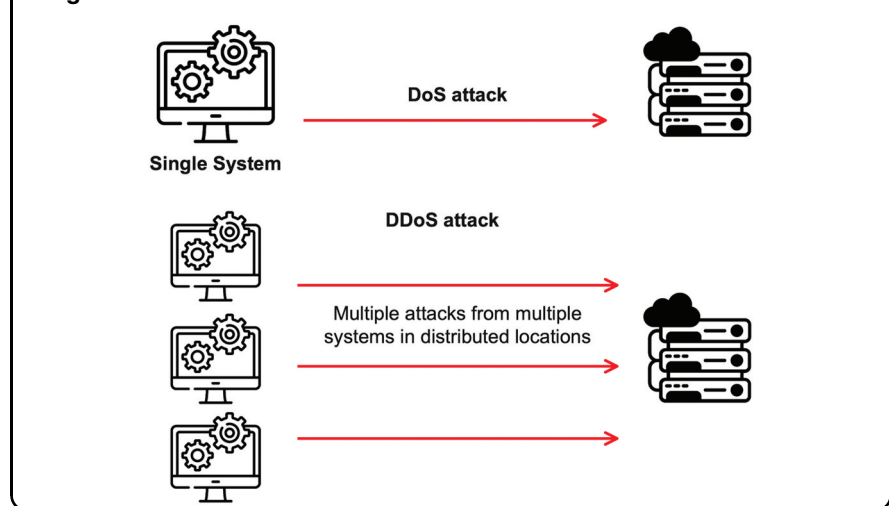
These assaults have the potential to cause serious operational interruptions, financial losses, and harm to the affected businesses' reputations. Attackers use a variety of techniques to boost malicious traffic, making mitigation attempts more difficult.

The number of searches to find a reliable and less-resource consuming solution is often increasing, while attacks are becoming more intelligent and complex (Bhushan & Sahoo, 2018).

Cryptography theoretically protects data by ensuring confidentiality, authenticity, and integrity, so that it is not accessible to just anyone, but it is not enough, and while DoS/DDoS attacks seek to make a machine or network resource unavailable, security based on cryptography could not be an efficient and practical solution for protecting a network from DoS attacks.

Recently, an intrusion detection system (IDS) has become a primordial component that aims to monitor and analyze traffic

Figure 1 DoS and DDoS attack.



in order to classify normal and abnormal status and can even take decision of prevention by issuing alerts. Machine learning is a science of programming machines like human do and without being directly guided. The procedure of learning is made through training and testing on specific data (Saranya et al., 2020).

Nowadays, security approaches and IDSs based on artificial intelligence and learning models have proven to be the most efficient and practical. They provide alternative solutions with low cost and high accuracy. Many studies have tested the accuracy and efficiency of different machine learning models for intrusion detection (Khedr et al., 2019; Maleh et al., 2021). Various learning models have demonstrated their efficiency such as random forest (RF), decision tree (DT), support vector machine (SVM), and so on.

Despite the importance of machine learning in protecting WSNs from vulnerabilities or malicious attacks, the accuracy of prediction depends two factors: the learning model used and the appropriate dataset used to obtain the highest rate and the approximation of the precision in the classification of attacks. This has led to the development of new approaches for accuracy enhancement. Ensemble learning, for example, is an approach of data mining that has an objective of performance and accuracy enhancement based on several mechanisms such as voting (majority or weighted), staging, and boosting.

This paper's primary objective is to conduct an analytical study of learning models in conjunction with ensemble learning techniques, with the aim of identifying a reliable security solution that accounts for the constrained resources of WSNs and extends the network's lifespan. Our approach centers on the application of classification algorithms, including RF, DT, and deep neural network (DNN), in order to achieve this goal. These algorithms were selected because prior research has shown them to be successful (Abbas et al., 2022). We have employed an ensemble learning strategy to increase the accuracy and overall performance of our suggested models.

In our contribution, we have especially focused on DoS, and DDoS attacks in WSNs and IoT, using two well-known benchmark datasets called CICIOT2023 and WSN-DS.

Our paper is organized as follows: Section "INTRODUCTION" introduces our contribution. Section "RELATED WORKS" reviews the related works that employed machine learning. Section "METHODOLOGY" presents the detailed methodology. Section "RESULTS AND DISCUSSION" presents the experiment and the results obtained from our approach. Finally, section "CONCLUSION" concludes our paper with suggestions for future work.

RELATED WORKS

Several studies have been published in recent years, providing several approaches. This section presents the latest research on DoS detection in WSNs using different learning models and various datasets.

One such study (Abbas et al., 2022) uses a WSN dataset of various sizes to assess the application of machine learning techniques on WSN nodes traffic and their impact on WSN network lifespan. They examined the performance measures of many machine learning models, such as multi-layer perceptrons (MLP), long short-term memory (LSTM), GBoost, DT, naïve Bayes, K-nearest neighbor (KNN), logistic regression (LR), and SVM. According to the experiment findings, the accuracy of the GBoost algorithm was 99.6%.

Another study (Yang et al., 2020) has proposed a scheme for intrusion detection in wireless networks. The authors have employed conditional deep belief network. The detection accuracy obtained for normal sample, flooding attack, false attack, and injection attack, respectively, are 98.9%, 80.8%, 72.7%, and 99.1%.

In another study, the authors (Sudar et al., 2021) used two machine learning models – DTs and SVM – and investigated DDoS attacks detection for training and testing, using the KDD99 dataset. The experimental results showed that the SVM algorithm outperformed the DT algorithm, with an accuracy rate of 85%.

Salmi and Oughdir (2023) developed and put into practice a number of IDS based on deep learning (DL), including CNN, DNN, RNN, and CNN+RNN. These models were assessed using industry-standard measures after being trained on the WSN-DS specialty dataset. The results of the experiment indicate that the CNN model, with an accuracy of 98.75%, performed best across all metrics.

A DNN-based detection system was proposed by Asad et al. (2020) to effectively detect DDoS attacks, which makes use of feed-forward back-propagation. With an accuracy rate of 98%, the suggested neural network architecture can recognize and make use of the most important high-level characteristics of packet flows.

A model based on a DL model and a contractive autoencoder to detect abnormal traffic was proposed by Aktar and Nur (2023). Three datasets, CIC-IDS2017, NSL-KDD, and CIC-DDoS2019, have been used to train the suggested model and to identify the normal traffic pattern from the compacted representation of the input data, and then they have applied a stochastic threshold method to detect the attack. The accuracy obtained varies according to each dataset, with the highest accuracy ranging between 93.41% and 97.58% on the CIC-DDoS2019 dataset.

For identifying intrusions, Patil and Pattewar (2022) have provided two methods: feature selection and majority vote. Majority voting is used in the design of an IDS with many models. The suggested method has been evaluated using a benchmark dataset from NSL-KDD. The experiment results indicate that using majority voting for the predicted models achieved the highest accuracy of 99.50% that was obtained using the chi-square features selection approach.

An approach to detect DoS malicious attacks was proposed by Otair et al. (2022), using the NSL-KDD dataset with two learning models: K-means and SVM. The accuracy obtained was 98.97%.

Anwer et al. (2021) analyzed the NSL-KDD dataset for DoS attacks detection using RF, and they achieved the highest accuracy rate of 85.34%.

Using three classification models and the SDN dataset, L. Barki et al. (2016) proposed an approach for DDoS detection that combines various feature selection mechanisms that reduce the dimension of the data and eliminate less significant features to enhance the prediction accuracy. All three models used, RF, SVM, and K-Nearest Neighbors, have demonstrated their efficiency, and the experiment results achieved an accuracy of 99.97%.

METHODOLOGY

This section introduces our research methodology, beginning with the dataset selected, the preprocessing step, and the metrics of evaluation chosen, in order to rigorously analyze the performance of three machine learning models to identify attacks in the networks, and finally the last step of evaluation using the ensemble learning approach.

Our approach can be summarized as follows as shown in Figure 2.

As summary of our contribution, as illustrated in Figure 2, we first loaded our two datasets separately, prepared each one, and split them into train and test. We evaluated our three models in classifying various types of attacks using various metrics of evaluation, as well as an ensemble learning approach is used to enhance the accuracy rate.

Dataset Selected

Within the field of intrusion detection simulation, the dataset utilized is derived from studies on information systems and sensor networks. In each IDS based on learning model, the choice of the dataset plays a paramount role to evaluate the performance of the model chosen. Generally, the dataset comprises both normal and abnormal traffic data, encompassing a limited range of attacks. For our experiments, we have opted to use two well-known benchmarks and real-time datasets, WSN-DS dataset (Almomani et al., 2016) and CICIOT2023 (Neto et al., 2023), due to their huge amount of data and diverse attack types.

CICIOT2023 is a dataset from the Canadian Institute for Cybersecurity (CIC) that contains various types of attacks divided into seven categories: DDoS, DoS, Recon, Web-based, Brute Force,

Figure 2 Model description.



Spoofing, and Mirai. While the Mirai attack system uses a huge DDoS assault to target IoT devices, the Recon attack uses the IoT topology to acquire detailed information about the target. Malicious IoT devices are the ones behind all of these assaults, which are directed against other IoT devices. CICIOT2023 dataset, created by Neto et al. (2023), is a cutting-edge and realistic collection of IoT assaults. It simulates the implementation of IoT goods and services in a smart home setting by utilizing a variety of topologies and 105 real IoT devices.

Almomani et al. (2016) developed WSN-DS dataset for intrusion detection in WSNs, particularly on four types of DoS attacks. This dataset is structured with 23 attributes generated by the LEACH protocol, featuring 374,661 records representing four DoS attack variations (Blackhole, Grayhole, Flooding, and Scheduling) alongside normal traffic. The selected dataset has the following distribution: Blackhole has 2043, flooding has 631, Grayhole has 2985 normal has 67,965, and TDMA has 1309.

Preprocessing

Before starting building our model, the preprocessing step is primordial; it significantly affects the quality of the models by enhancing the level of accuracy. To carry out this step, many tasks must be done, including:

- **Removing spaces from the column names**

When the column names contain spaces at the beginning and the end of the names, this will cause problems when we try to access the columns by their names.

- **Removing irrelevant columns**

Some columns are not useful for our analysis and may cause problems when we try to build our models, so we need to remove these columns.

- **Removing the rows that contain missing values**

Missing values in the dataset may cause problems when we try to build our models, so we need to remove the rows that contain missing values.

- **Removing duplicates**

Duplicates in the dataset may cause overfitting when we try to build our models, so we need to remove the duplicates.

- **Label mapping**

This step includes converting categorical labels into numerical values suitable for machine learning algorithms.

- **Data balancing**

When we found that the dataset was imbalanced, we needed to balance it by upsampling the minority classes and downsampling the majority classes, as seen in Figures 3 and 4.

- **Features scaling**

Normalizing the data ensures that every feature adds the same amount to the model's learning. This often involves scaling

features to a standard range, such as [0, 1], and applying transformations to stabilize variance and enhance model performance.

- **Data splitting into training and testing**

We divide the data, to train our model on the training set, and the rest of the data will be used for the model test, to evaluate the accuracy. This splitting of data into training and testing prevents the overfitting problem.

Model Building

- **DT model**

DT is a well-known supervised learning model, with hierarchical representation, composed of root node, which represents the entire dataset; leaf node, which is the final output node; decision nodes; and sub-trees. In a DT, the data is continuously split based on a specific parameter and the most significant feature (Lee et al., 2008). This learning model is strongly recommended for classification problems when the data is not linearly separable. According to various studies, the DT is a learning model that is expected to be a suitable solution for IDS, especially the anomaly detection (Alotaibi & Elleithy, 2016).

The DT model was built using the `DecisionTreeClassifier` class from the `sklearn.tree` module.

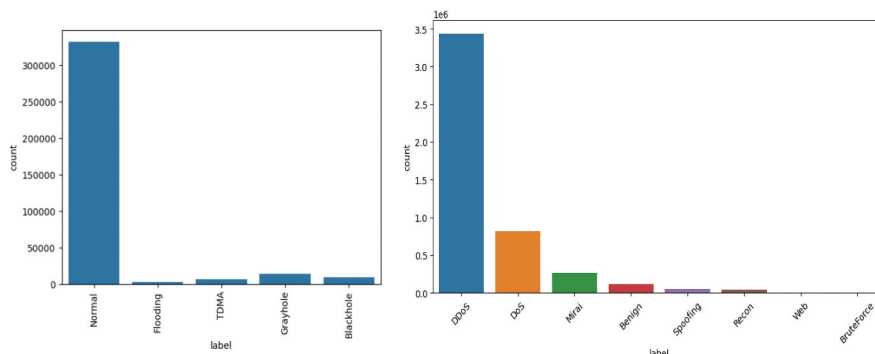
- **RF model**

Random Forest (RF) is a supervised learning model that can be used for both classification and regression. Its main principle involves creating large trees during the training step. Each tree in the forest makes its own decision, and the final output prediction of the forest is the class chosen by the majority of trees, as illustrated in Figures 3 and 5. The more trees used, the better the accuracy achieved.

Generally, we can summarize the RF process into three main steps:

- Construction of n DTs by randomly drawing n samples of observations,
- Training each DT,

Figure 3 *WSN-DS and CICIoT2023 datasets before balancing.*



- To make a forecast on new data, it is necessary to apply each of the n trees and take the majority among the n forecasts.

The RF model was built using the RandomForestClassifier class from the sklearn.Ensemble module.

• Deep Neural Network

DL is a subfield of machine learning based on artificial neural network classification, destined for complex data. The concept DL is inspired by the human brain, in which ways to represent data with multiple layers as seen in Figure 6, in order to extract significant features, to be able to reach the optimal result (Q. Cao et al., 2018).

Figure 4 WSN-DS and CICIoT2023 datasets after balancing.

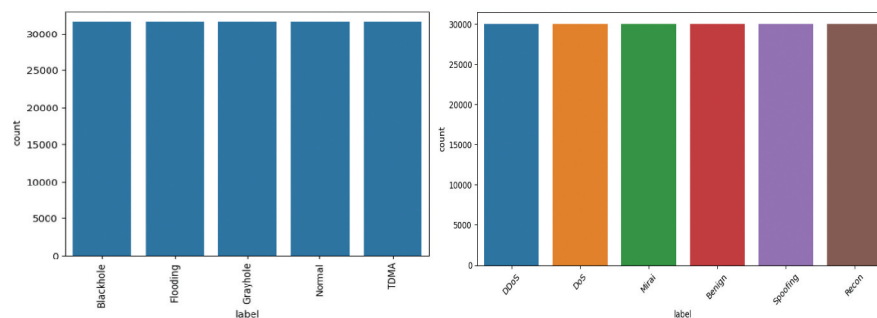


Figure 5 Random forest model.

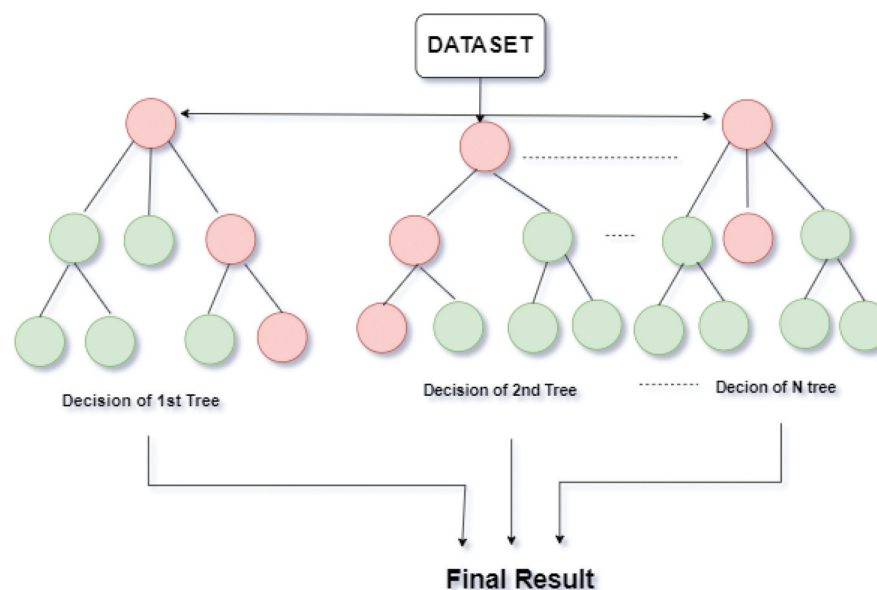
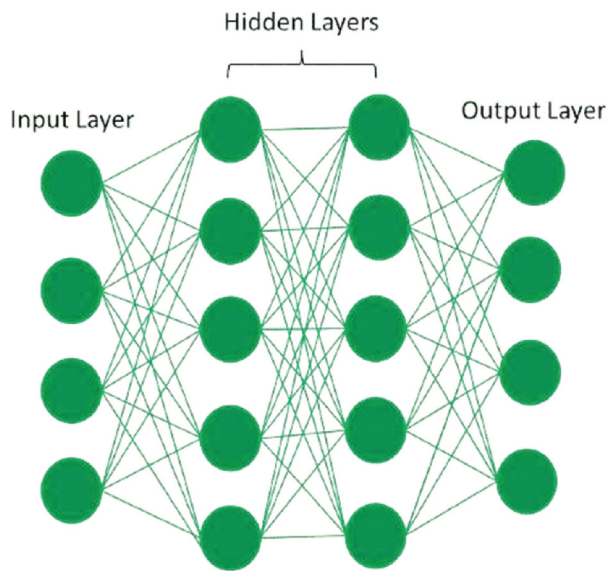


Figure 6 DL architecture.



According to many recent studies, many areas in machine learning, such as object detection, voice and image recognition, and language translation, have been significantly impacted and improved by the development of DL (Abbas et al., 2022; Dahl et al., 2011; Girshick et al., 2015).

Despite the advantage of DL and its ability to function with or without labels, DL approach usually requires high computational power and dedicated GPUs to work and to achieve good results.

DNN becomes a vessel of knowledge, traversing through the intricacies of data, illuminating its true essence, and capturing the essence of complexity within its expanding reaches. In the grand tapestry of computation, the DNN stands tall as an emblem of human ingenuity, a testament to our ability to decode the enigmatic realm of intelligence, and a beacon guiding us toward a future where the boundaries of human knowledge are pushed ever further (Parkhi et al., 2015). Figures 5, 7 and 8 present the DNN architecture implemented in this work:

Evaluation Metrics

In machine learning (ML), evaluation metrics are crucial for assessing the performance of a model and understanding how well it generalizes to unseen data. The choice of evaluation metrics depends on the type of problem you are solving (classification, regression, etc.). With TP denoting True Positives, TN representing True Negatives, FP signifying False Positives, and FN indicating False Negatives, the evaluation metrics employed in this study are respectively:

- Confusion matrix: A table represent the summary of correct and incorrect predictions using the indicators TP, TN, FP, and FN (Table 1).

- Accuracy: in the context of this research, it is a metric that gauges the overall correctness of the model. It is calculated using the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Figure 7 DNN architecture implemented in WSN-DS dataset.

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_6 (Dense)	(None, 100)	1900
dense_7 (Dense)	(None, 100)	10100
dense_8 (Dense)	(None, 5)	505

Total params: 12505 (48.85 KB)

Trainable params: 12505 (48.85 KB)

Non-trainable params: 0 (0.00 Byte)

Figure 8 DNN architecture implemented in CICIoT2023 dataset.

dnn_model.summary()

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_9 (Dense)	(None, 46, 256)	512
dropout_6 (Dropout)	(None, 46, 256)	0
dense_10 (Dense)	(None, 46, 256)	65,792
dropout_7 (Dropout)	(None, 46, 256)	0
flatten_1 (Flatten)	(None, 11776)	0
dense_11 (Dense)	(None, 6)	70,662

Total params: 136,966 (535.02 KB)

Trainable params: 136,966 (535.02 KB)

Non-trainable params: 0 (0.00 B)

Table 1 *Confusion matrix representation.*

		Predicted Class	
		Normal	Attack
Actual class	Normal	True Positive	False Negative
	Attack	False Positive	True Negative

- Recall: in the context of this research, it is a metric that assesses the model's ability to capture all the relevant instances. It is calculated using the formula:

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

- Precision: the ratio of correctly identified labels to the total number of positive classification

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

- F1-Score: in the context of this research, it is a metric that balances precision and recall into a single measure. It is calculated using the formula:

$$F1 - score = \frac{2TP}{2TP + FN + FP} \quad (4)$$

Ensemble Learning

Ensemble learning is a technique that combines the predictions of multiple classification models to improve the accuracy rate of prediction results (Ennaji et al., 2024).

The basic idea of ensemble learning systems is that the prediction errors made by one model can be compensated by the correct predictions of another model and so on.

Several categorizations of ensemble learning methods can be defined. They can be presented by basic methods like voting (majority or weighted voting) and advanced methods like stacking, bagging, and boosting (Ennaji et al., 2024).

In our contribution, we have used the voting majority method (Alotaibi & Elleithy, 2016), which is a simple aggregation approach, based on aggregating predictions from multiple base models (Figures 4 and 9

Majority voting is used in classification, where each base model gives a prediction, or the class that gets the highest number of votes is chosen as the final prediction, defined by the following relation:

$$, Final Prediction = \operatorname{argmax} \sum Vote(i) \quad (5)$$

where vote (i) is the prediction of model i.

RESULTS AND DISCUSSION

In this section, we present the experiment result of each model before and after applying the ensemble learning approach, as well as an analysis comparison with the previous works.

The proposed models were implemented within Google Colab, on an Intel CORE i5, 16 GB ram, in Windows 10 operating system. To run our models, we have used various Python packages such as Numpy, Pandas, Sklearn, and so on.

Experiment Result

The assessment of the research findings was carried out meticulously, employing a variety of measures to measure the effectiveness of the three models in detecting DoS attacks, particularly in WSNs.

• DT result using the WSN-DS and CICIOT2023 datasets

As a summary of DT result, we provide a confusion matrix and an accuracy report as seen in [Figures 10, 11 and 12](#) for more details.

The experiment results obtained from the DT model demonstrate its effectiveness with high accuracy achieved in classifying two types of DoS (Blackhole and Grayhole) and DDoS attacks for both datasets.

• RF result using WSN-DS and CICIOT2023 datasets

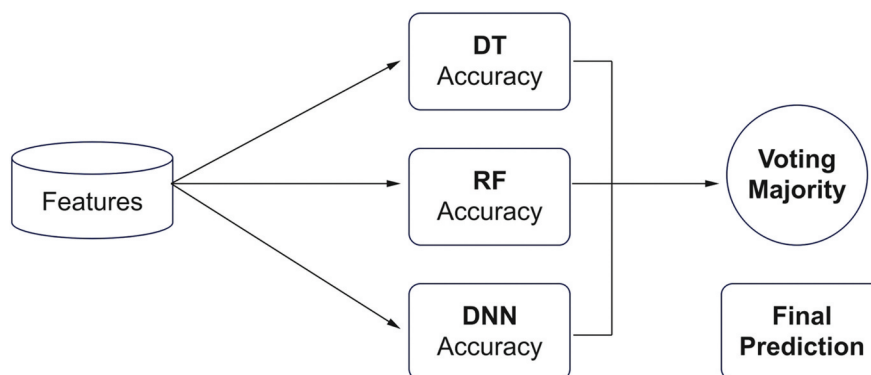
As a summary of RF result, we provide a confusion matrix and an accuracy report, as illustrated in [Figures 13, 14 and 15](#).

According to the experiment result, the RF indicates a good performance of the model in classifying Blackhole and TDMA attacks in the WSN-DS dataset and DoS and DDoS in CICIOT2023 dataset.

• DNN result

- the experiment result of DNN model is illustrated in [Figures 16, 17 and 18](#)

Figure 9 Voting majority method.



- **Voting majority result**
- the experiment result of the voting majority is provided in [Figure 19](#)

According to the classification report of the three tested models ([Figures 10, 11, 12, 13, 14, 15, 16 and 17](#)), the results obtained have demonstrated impressive performance, with accuracy, precision, and recall. After using the ensemble learning based on majority voting, the accuracy is remarkably improved ([Figure 19](#)), in comparison with each model separately and in comparison with the related works ([Table 2](#)).

Figure 10 Accuracy report and confusion matrix of decision tree using WSN-DS dataset.

Classification Report:										
	precision	recall	f1-score	support						
Blackhole	0.99	0.99	0.99	2043						
Flooding	0.95	0.94	0.95	631						
Grayhole	0.98	0.98	0.98	2985						
Normal	1.00	1.00	1.00	67965						
TDMA	0.92	0.94	0.93	1309						
accuracy			0.99	74933						
macro avg	0.97	0.97	0.97	74933						
weighted avg	0.99	0.99	0.99	74933						

Confusion Matrix:					
[[	2031	0	12	0	0]
[	0	618	0	13	0]
[	19	0	2942	24	0]
[	0	40	35	67885	5]
[	3	0	1	84	1221]]

Figure 11 Accuracy report of decision tree using CICIoT2023 dataset.

Decision Tree Accuracy: 0.9256666666666666				
Decision Tree Classification Report:				
	precision	recall	f1-score	support
Benign	0.78	0.93	0.85	5933
DDoS	1.00	0.98	0.99	5976
DoS	0.99	0.99	0.99	5922
Mirai	1.00	0.99	1.00	6086
Recon	0.87	0.88	0.88	6119
Spoofing	0.96	0.77	0.85	5964
accuracy			0.93	36000
macro avg	0.93	0.93	0.93	36000
weighted avg	0.93	0.93	0.93	36000

Discussion

After evaluating our three models, we have found that this approach has high accuracy, precision, recall, and f1-score.

This indicates that the model correctly classified traffic, whether normal or attack and that the high precision achieved suggests that when the model classified traffic as a DoS or DDoS attack, it was correct.

In summary, our work provides several advantages:

Figure 12 Confusion matrix of decision tree model using CICIoT2023 dataset.

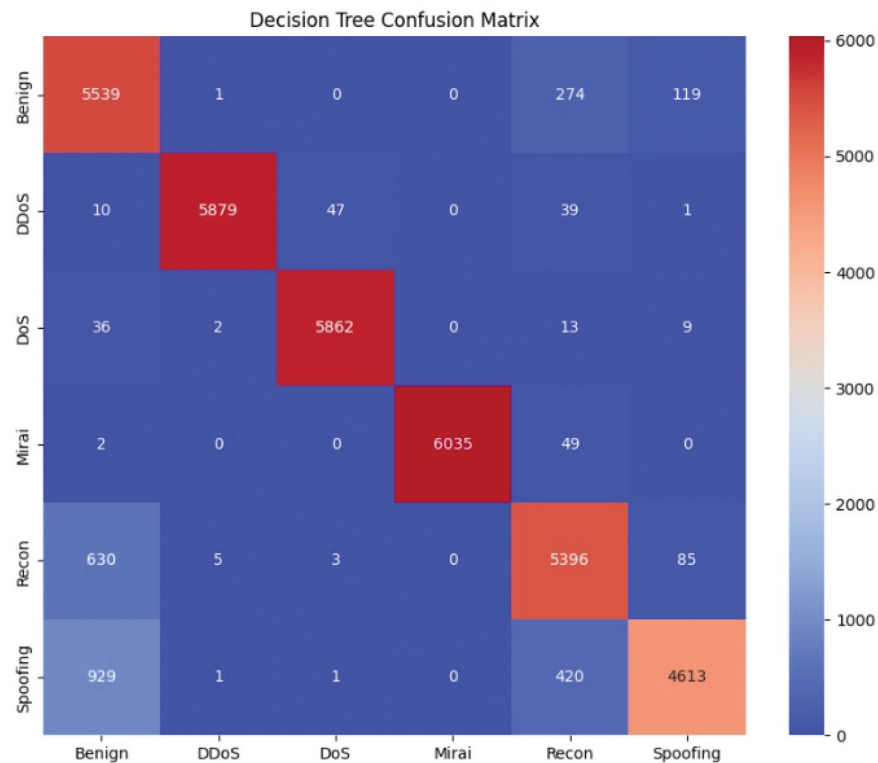


Figure 13 Accuracy report and confusion matrix of random forest using WSN-DS dataset.

Classification Report:					Confusion Matrix:
	precision	recall	f1-score	support	
Blackhole	0.99	0.99	0.99	2043	
Flooding	0.94	0.98	0.96	631	
Grayhole	0.98	0.99	0.98	2985	
Normal	1.00	1.00	1.00	67965	
TDMA	1.00	0.93	0.96	1309	
accuracy			1.00	74933	
macro avg	0.98	0.98	0.98	74933	
weighted avg	1.00	1.00	1.00	74933	
					[[2031 0 12 0 0]
					[0 618 0 13 0]
					[19 0 2942 24 0]
					[0 40 35 67885 5]
					[3 0 1 84 1221]]

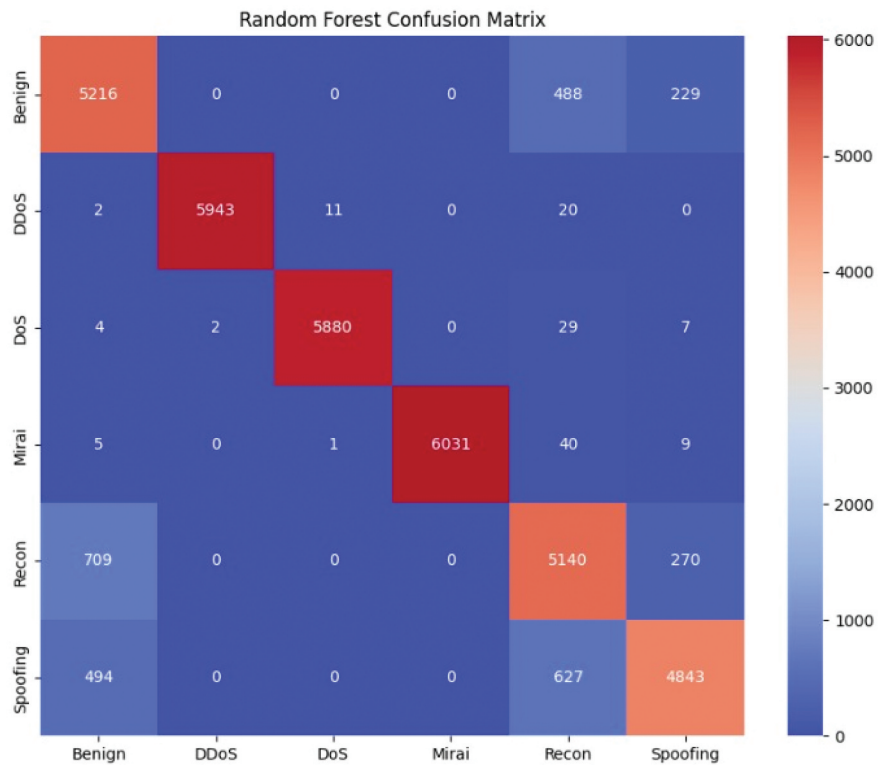
Figure 14 Accuracy report of random forest using CICIoT2023 dataset.

Random Forest Accuracy: 0.9181388888888888

Random Forest Classification Report:

	precision	recall	f1-score	support
Benign	0.81	0.88	0.84	5933
DDoS	1.00	0.99	1.00	5976
DoS	1.00	0.99	1.00	5922
Mirai	1.00	0.99	1.00	6086
Recon	0.81	0.84	0.82	6119
Spoofing	0.90	0.81	0.86	5964
accuracy			0.92	36000
macro avg	0.92	0.92	0.92	36000
weighted avg	0.92	0.92	0.92	36000

Figure 15 Confusion matrix of random forest model using CICIoT2023 dataset.



- **High Accuracy:** Achieves superior performance in detecting a wide range of network intrusions, including DoS and DDoS attacks.
- **Scalability:** The model's architecture allows it to scale effectively with increasing data volumes, making it suitable for real-time intrusion detection.
- **Resource Efficiency:** Optimized for deployment in resource-constrained environments such as IoT and WSNs, where computational and energy resources are limited.

Figure 16 Accuracy report of DNN using WSN-DS dataset.

Classification Report:				
	precision	recall	f1-score	support
Blackhole	0.78	0.88	0.83	2043
Flooding	0.90	1.00	0.95	631
Grayhole	0.90	0.82	0.86	2985
Normal	1.00	1.00	1.00	67965
TDMA	0.99	0.93	0.96	1309
accuracy			0.99	74933
macro avg	0.91	0.93	0.92	74933
weighted avg	0.99	0.99	0.99	74933

Figure 17 Accuracy report of DNN using CICIoT2023 dataset.

DNN Accuracy: 0.8669722222222223				
DNN Classification Report:				
	precision	recall	f1-score	support
Benign	0.75	0.72	0.73	5933
DDoS	1.00	0.98	0.99	5976
DoS	0.98	1.00	0.99	5922
Mirai	1.00	1.00	1.00	6086
Recon	0.73	0.81	0.77	6119
Spoofing	0.76	0.70	0.73	5964
accuracy			0.87	36000
macro avg	0.87	0.87	0.87	36000
weighted avg	0.87	0.87	0.87	36000

These advancements highlight the potential of ensemble learning approach to significantly enhance the robustness and reliability of IDS in protecting modern network infrastructures against evolving cyberthreats.

Despite the effectiveness achieved in our approach, there are some limitations. As attackers enhance their behaviors, our approach still needs to be tested rigorously with other learning models, using recent datasets with various types of attacks. The limited resources of WSNs are always a challenge that must be

Figure 18 Confusion matrix of DNN using CICIoT2023 dataset.

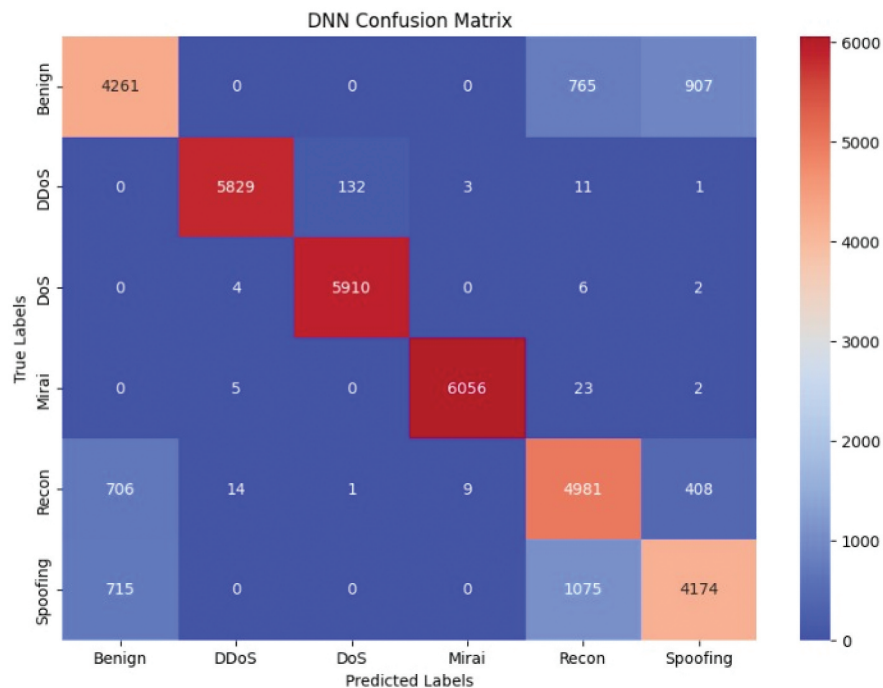


Figure 19 Accuracy report and confusion matrix of voting majority using WSN-DS dataset.

Accuracy: 1.00					[[2022 0 21 0 0] [0 608 0 23 0] [22 0 2936 25 2] [0 35 27 67897 6] [3 0 1 82 1223]]
	precision	recall	f1-score	support	
0	0.99	0.99	0.99	2043	
1	0.95	0.96	0.95	631	
2	0.98	0.98	0.98	2985	
3	1.00	1.00	1.00	67965	
4	0.99	0.93	0.96	1309	
accuracy			1.00	74933	
macro avg	0.98	0.97	0.98	74933	
weighted avg	1.00	1.00	1.00	74933	

Table 2 Comparative analysis.

Reference	Methods and Approach	Accuracy Output	Reference	Methods and Approach	High Accuracy
Abbas et al., (2022)	KNN, LR, Gboost, SVM, DT, NB, MLP WSN-Dataset	99.6%	S. Aktar et al., (2023)	Feature Selection Majority voting	99.5%
Sudar et al., (2021)	DT, SVM Dataset: KDD99	85%	Otair et al., (2022)	KNN, SVM Dataset : NSL-KDD	98.97%
Salmi et al., (2023)	DNN, CNN, RNN, CNN +RNN Dataset: WSN-DS	98.75%	Anwer et al., (2021)	RF Dataset: NSL-KDD	85.34%
Asad et al., (2020)	Deep Neural Network	89%	Barki et al., (2016)	RF, SVM, KNN SDN-Dataset	99.97%
Aktar et al., (2023)	Deep learning 3 Datasets: NSL-KDD/ CICDDoS2017:2019	93.41% <X<97.58%	Our contribution	DNN, DT, RF with Voting majority 2 Dataset: WSN-DS, CICIOT2023	Majority voting with an accuracy of 100%

taken into consideration when developing a security solution with high accuracy and minimum execution time.

CONCLUSION

Several security approaches focus to make WSNs secure, such as authentication, encryption, key management, etc. Unfortunately, not all of these mechanisms are enough. Recently, machine learning has provided an alternative mechanism of security with low cost, by decreasing the energy and increasing the lifetime of WSNs. In this paper, we have focused on DoS detection using a well-known dataset WSN-DS. We have evaluated three learning models, DL model (DNN), DT and RF, using the standard metrics of evaluation. We have also integrated an approach ensemble learning based on majority voting, and the result obtained demonstrate its effectiveness as an intrusion detection method.

In future work, we will use multiple datasets to evaluate other learning models as a hybrid approach. We think about using other advanced ensemble learning methods such as stacking, boosting, or bagging, which have recently shown improvements in classification accuracy.

DISCLOSURE STATEMENT

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